

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DIGITAL SIGNAL PROCESSING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Use of Non – programmable calculator is allowed.
- (3) Assume suitable data wherever necessary.

1. (a) Reconstruct and plot the causal signal $x(n)$ from its DFT given by,
 $X(k) = \{1.42, -0.88 - j2.30, 0.33, -0.88 + j2.30\}$ 5(CO1,2)
(b) Compute 8-Point DFT of sequence $x(n) = \{1, 0, 0, 1, 1, 1\}$. Use Radix-2,
DIF-FFT Algorithm. 5(CO1,2)
2. (a) Realize the system given by difference equation –
 $y(n) = -0.1 y(n - 1) + 0.2 y(n - 2) + 3x(n) + 3.6 x(n - 1) + 0.6 x(n - 2)$ in
(1) Direct form-II.
(2) Cascade form. 5(CO1,3)
(b) Realize the given system function using minimum number of multipliers :
 $H(z) = 2 + 0.75z^{-1} + 0.25z^{-2} + 0.75z^{-3} + 0.25z^{-4} + 0.75z^{-5} + 2z^{-6}$
5(CO1,3)
3. Design digital IIR Butterworth LPF filter using bilinear transformation with following
Specifications :
Pass band ripple 3 dB up to 2π rad/Sec.
Stop band attenuation 20 dB beyond 8π rad/Sec.
Sampling frequency 500 Hz. 10(CO1,3)

4. (a) Design a causal linear phase FIR filter by Hamming window for following specifications :
- $$H_a(f) = 1 \text{ for } 0 \leq f \leq 1000 \text{ Hz}$$
- $$= 0 \text{ for } f > 1000 \text{ Hz}$$
- Sampling frequency 10000 Hz and duration of impulse response of filter is 1 msec. (Hint. : Length of filter should be calculated from duration of impulse response). Also realize the causal linear phase filter structure with minimum multiplier. Use Hamming Window given by –
- $$W_H(n) = 0.54 - 0.46 \cos(2\pi n/(N-1)), \quad 0 \leq n \leq N-1 \quad 8(\text{CO1},3)$$
- (b) Distinguish between FIR and IIR filters. 2(CO1,3)
5. (a) Implement a two stage decimator for the following specifications :
- Sampling rate of the input signal = 20000 Hz
- $M = 100$, Passband = 10 to 40 Hz
- Transition band = 40 to 50 Hz
- Passband ripple = 0.01
- Stopband ripple = 0.002 5(CO1,4)
- (b) What are Quadrature mirror filter banks ? Give input-output analysis for a two channel QMF filter bank. 5(CO1,4)
6. (a) What are the on chip peripherals available on programmable digital signal processors ? Explain their functions with block diagram. 5(CO1,5)
- (b) State the advantages and applications of DSP processors. 5(CO1,5)

